Client B (Structural Steel): from "AI for nesting" to a measured material-yield programme

Prepared by Banaov Technologies, Forward Deployed Engineering. Based on two discovery calls. Nothing has been proposed to the client yet. All figures in USD.

1. Summary

The client asked where AI can help and named nesting first. After two calls my reading is that their real problem is not a missing optimiser. They cannot say how much steel they waste, how much of it was avoidable, or what surplus material is sitting in the yard. Until that is measured, nobody can tell whether an AI nesting engine would help, and the one team that knows (the nesters) was not on the calls and disputes the premise.

I recommend a 20-week fixed-price programme with a go/stop gate at week 8:

1. A waste ledger built from the data they already hold in Strumis, so cost control can see waste by project, material and cause.
2. A benchmark that replays historical nests: manual vs Strumis automatic nesting vs our logic. This settles the "manual saves more" claim with data.
3. A remnant and surplus matcher that suggests existing offcuts and other-project surplus before new steel is bought or nested. A nester accepts or rejects every suggestion.

Price $165,000, base-case value about $308,000 a year, first visible result (the baseline waste report) in week 4. I would not build a replacement nesting engine, a yard optimiser or a trailer load planner first.

2. Problem understanding

2.1 What they asked for vs what they need

They asked for	What the notes show they need
"AI to optimise nesting"	Measurement first. They have "manual data" saying plate waste is 20 to 25%, but no way to split it into unavoidable vs avoidable, or to see unused stock.
A consultation, then development	A decision they can defend to head office, with evidence, in weeks not months. The contact must give an update "within this week".
Automation of allocation	A rule change as much as software. Material cost is loaded to one project at purchase, so using project A's surplus on project B needs a cost-transfer rule from finance. The contact himself is unsure it is feasible.
Help across five or six areas	A ranked order. Areas differ hugely in data readiness and in who must cooperate.

Things in the notes that do not add up, and how I treated them:

- Plate waste 20 to 25%, profiles 5 to 10%, but overall 7 to 10%. A tonnage-weighted blend of those two ranges only reaches 7 to 10% if plate is roughly 0 to 17% of tonnes (Appendix B). Either plate is a small share of the steel, or the plate figure is not measured the same way as the overall figure. This is my first question for the next call. It matters because if profiles are 85%+ of tonnes, a point of profile yield is worth more than plate.
- "We don't have the data" vs "the manual data shows 20 to 25%". I read this as: records exist, but are not trusted or comparable (revisions untracked, offcuts recorded but not reused).
- The nesting team says automatic nesting wastes more. The client is not sure. It may be true (nesters credit offcuts they plan to reuse, auto nesting may ignore them), or it may be a parameter or trust issue. I treat it as a hypothesis to test, not a fact.

2.2 Stakeholders

Who	What they want	Concern
Cost Control engineer (our contact)	Visibility of waste and loss; claim-back for client-caused revisions; something to report to head office	Does not know nesting; cannot commit other teams
Management / head office	Lower material cost; fewer hired-in trailers (they are "very unhappy")	Wants proof before spending
Nesting team	To keep nest quality and ownership; credit for offcut planning	Being judged by a tool or replaced; they believe manual beats auto
Engineering	Fewer disputes about revisions	Being blamed for "engineering errors" that nobody quantifies
Purchasing and planning	Buy to project timelines, hit delivery dates	Early buying drives surplus and yard clutter
Stores / yard	Find material fast	Plates buried at the bottom, no storage sequence
Finance	Clean per-project cost loading	Cross-project transfers breaking project margins
Factory (sizing/CNC), dispatch, fleet	Nest files that run; trailers fully used	Not discussed in depth

Also relevant: another supplier has proposed buying software packages. Our difference is that we build and deliver with our own team.

2.3 As-is process (as I understand it)

Business development bids, estimation prices it. On a win, planning schedules procurement and fabrication and promises a completion date; engineering releases fabrication drawings (revisions keep arriving, with no clean record). Purchasing buys full-size plate and sections per project. The nesting team nests manually in Strumis, checking offcuts first; leftovers are logged as offcuts. The CNC factory then does sizing, fabrication (fit-up, welding), coating (paint, galvanising, fireproofing), and dispatch ships it, using about 13 own trailers plus hired ones.

Pain points: no waste analytics; offcuts underused; purchase is project-specific, so surplus cannot flow between projects; drawing revisions untracked; storage is not sequenced; many projects currently on hold, so plans keep changing.

3. What we don't know yet

Ranked by how much each answer could change the plan. Next call should be 60 to 90 minutes, ideally with a nesting lead and

someone from stores.

#	Question	Assumption until answered
1	What share of tonnes is plate vs profiles vs other, and how is "waste %" computed (denominator, are returned offcuts counted as waste)?	Plate about 10% of tonnes, profiles about 85%, other 5%. Overall waste 8.5%.
2	Can we get read-only access to Strumis data (purchases, nests, offcuts, stock, drawing releases and revisions), by export or database? Who administers it?	CSV or SQL read access within 2 weeks. If not, add 2 weeks and I scope manual exports.
3	May material bought for one project be used on another? Commercially, contractually (mill certificates, heat-number traceability), and in cost loading?	Allowed for non-critical grades when the receiving project's spec accepts it, with a cost-transfer journal approved by Cost Control. Otherwise phase 2 stays within one project.
4	Why does the nesting team prefer manual nesting? Can we sit with them and get 3 to 5 real jobs where both methods can be compared?	Their claim is partly true because offcut reuse and local rules are not captured in auto settings.
5	How are offcuts recorded: dimensions, grade, thickness, heat number, physical location, tagged?	Dimensions, grade and thickness recorded; location unreliable; some offcuts never logged.
6	How much waste is caused by revisions after purchase or nesting, vs our own engineering errors, vs nesting loss?	Revisions are a large cause (guess 20 to 30% of avoidable waste) but untracked, so phase 1 only captures them as a cause code.
7	Who decides: who approves budget, who owns nesting rules, how big is the nesting team?	Contact sponsors; head office approves; 4 to 6 nesters with one lead.
8	What is the steel price paid and the scrap resale price? What currency and region?	$800/t purchase, $250/t scrap, so net loss $550 per tonne wasted.
9	How many projects are currently on hold, and how much new drawing volume will exist over the next 6 months?	10 to 15 active, with enough new drawings to pilot on 2 to 3.
10	IT: where can data and a tool live (their servers, their cloud tenant, ours), and what security review is needed?	Their cloud tenant or a server on site; read-only data feed.

4. Solution

4.1 Options considered

Option	Value	Data ready?	Needs other teams?	Time to first result	Verdict
A. Switch on or buy a commercial nesting optimiser	Medium	Partly	Nesters must adopt	Fast	Possible outcome of the benchmark, not a first step; the nesters already reject Strumis auto nesting
B. Custom AI nesting engine	Uncertain	No	Yes	Slow	Not first. Cannot be justified without baseline
C. Waste ledger + benchmark + remnant/surplus matcher	High	Mostly (Strumis)	Nesters, stores, finance	4 weeks	Recommended
D. Drawing revision tracking	Medium to high	Poor	Engineering	Medium	Phase 2 add-on; cause code captured in C
E. Yard storage re-slotting	Medium	Poor (locations unreliable)	Stores	Medium	Later; depends on C providing stock records
F. Trailer load planning	Medium	Unknown	Dispatch	Medium	Later; own fleet is only about 13 trailers, the bigger lever is utilisation
G. Fleet utilisation diagnostic (CarTrack + booking data)	Medium	Yes, six months collected	Fleet	2 to 3 weeks	Optional parallel quick win for management ($12,000)

4.2 Recommendation: option C

Why it fits this client:

- It is their own first pick, but it respects what they said they lack: data. The ledger is useful even if everything else is stopped.
- It uses data they already have, so the risk of waiting on other teams is lower than for D, E or F.
- The benchmark protects them from paying for the wrong thing, including from us. If auto nesting or a commercial package wins, we say so.
- The remnant matcher attacks waste that no optimiser can fix: steel already bought and sitting unused.
- It is decision support. A nester signs off every plan, which handles the adoption risk.

What I would deliberately not build: our own nesting engine to replace Strumis; any write-back into Strumis in this phase; an LLM chatbot over their ERP; automated purchasing; a yard digital twin; a trailer load optimiser; a new ERP or dashboards for every department.

5. Solution engineering

Honest framing: this is mostly data engineering and operations research (cutting-stock and bin-packing heuristics), not machine learning. ML can come later for demand forecasting if the data justifies it.

5.1 Inputs and sources

All from Strumis unless noted, read-only, nightly extract:

- Purchase orders and goods received: material type, grade, size, heat number, project, date, mass.
- Nest results: nest ID, plate or length used, parts nested, yield, nester, date.
- Offcut and remnant records: dimensions, grade, thickness or section, project of origin, location, age.
- Drawing releases and revisions: date, project, revision number.
- Project cost loading rules from Cost Control (spreadsheet or Strumis).
- Rules from the nesting team: minimum usable offcut size, kerf, grain/orientation, grade substitution rules.

5.2 Components

1. Waste ledger. A mass balance per project and material class: purchased = in finished parts + usable offcuts returned to stock + scrap + unexplained. Each loss gets a cause code: nesting loss, profile end drop, unused offcut aged more than 90 days, revision,

defect, engineering error. Unexplained is shown, not hidden. Output: a web dashboard and a weekly PDF for Cost Control.
2. Benchmark harness. Replays historical nest jobs with identical part lists through (a) the manual result as recorded, (b) Strumis automatic nesting, (c) our logic: plate nesting using an open-source nesting library with offcut reuse; profiles with a 1D cutting-stock solver (OR-Tools). Compares yield and time.
3. Remnant and surplus matcher. When drawings are released, or before a purchase order is placed, it takes the part list and does:
   - Filter stock by grade, thickness or section, traceability class, and project permission.
   - Pack parts into candidate offcuts or surplus items, own project first, then other projects.
   - Rank suggestions by tonnes of new purchase avoided.
   - Show a "net requirement after stock" figure for Purchasing.

5.3 Where a person signs off

- The nester accepts or rejects each suggestion and gives a reason code. Rejections feed the rules.
- Stores confirms the physical piece exists and is findable before it is reserved.
- Cost Control approves any cross-project transfer and the cost-transfer journal.
- The nesting lead owns the rule set (minimum offcut size, kerf, substitutions). We never change it silently.

5.4 Fit with existing systems, people and routine

- Read-only from Strumis. Suggestions leave as a PDF or CSV the nester applies in Strumis, so no integration risk to production nesting.
- It runs automatically when drawings are released and daily on stock; nesters use a single screen, no new login steps beyond that.
- A 30-minute weekly review with Cost Control, nesting lead and stores looks at the ledger and the rejected suggestions.
- Optional later: write-back, and ingesting CarTrack and booking data for option G.

6. Evaluation

Baseline. Twelve months of Strumis history (or what exists), reconstructed per project and material class, using the ledger definitions agreed in week 2. If the client's 7 to 10% overall figure does not match our reconstruction, we report both and agree one definition in writing.

Primary metric. Material yield: finished-part mass divided by net purchased mass, where an offcut returned to stock counts as recovered only if it is used within 90 days. Reported separately for plate and profiles, and overall.

Secondary metrics. Offcut utilisation rate (offcut mass consumed divided by offcut mass created within 90 days); share of suggestions accepted; nester time per nest; tonnes of purchase avoided.

Test protocol.

1. Back-test (weeks 4 to 8): at least 40 historical nest jobs or 3 months of projects, frozen before we look at results, compared against the manual result actually achieved.
2. Shadow mode (weeks 11 to 12): the tool suggests, nesters ignore it, we compute the counterfactual.
3. Live pilot (weeks 13 to 16): 2 to 3 projects, nesters can accept, compared with matched earlier projects (same material class and complexity) and with the shadow estimate.

Thresholds. Success means at least 0.5 percentage points of overall yield improvement (base case in the value model is 0.7), or at least 3 points on plate if plate turns out to be the bigger share of the loss, plus at least 60% of suggestions accepted.

Stop conditions.

- Week 4: we cannot reconstruct the mass balance for at least 70% of projects, so the programme is a data-quality project and we say so.
- Week 8 gate: the back-test shows less than 0.3 points improvement overall, or the surplus matched is under 5% of the tonnes needed. We hand over the ledger and benchmark and stop.
- Any time: Finance refuses cross-project transfers and in-project reuse alone gives under 0.2 points, or nesters reject over 70% of suggestions for reasons we cannot fix.

7. Build plan and timeline (20 weeks)

Phase	Weeks	What happens	Milestone
0 Onboard	1 to 3	Site visit; meet nesting, stores, engineering, finance; get data access; agree definitions	M0 kickoff (wk 1)
0b Baseline	3 to 4	First waste ledger on historical data	M1 baseline report (wk 4)
1 Benchmark	4 to 8	Back-test manual vs auto vs our logic; design matcher rules with nesters	M2 gate: go or stop (wk 8)
2 Build and pilot	9 to 16	Matcher, dashboard, PO-time net requirement; shadow, then live pilot	M3 pilot live (wk 16)
3 Roll out	17 to 20	All new projects, training, runbook, measured results report	M4 acceptance (wk 20)

Depends on the client: Strumis read access by end of week 2; named nesting lead, stores contact and Finance contact with about 2 hours a week each; a decision on cross-project transfers by week 6; a nester to validate suggestions during the pilot; agreement on metric definitions in week 2; quick security or IT approval.

If data access slips, every phase moves by the same number of weeks and we say so at the next weekly check-in.

8. Cost

8.1 Our effort and team

Role	Weeks	Rate per week	Cost
FDE lead (owner, onsite, stakeholder)	20	$2,400	$48,000
Data/ML engineer	18	$2,000	$36,000
Full-stack and integration	8	$1,800	$14,400
PM and QA (0.2 FTE)	4	$1,800	$7,200
Labour subtotal	50 person-weeks		$105,600
Travel and accommodation (3 trips at $2,500)			$7,500
Contingency (10% of labour)			$10,560

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Total delivery cost			$123,660

8.2 Price and payment

Fixed price 165,000, *whichisa*2541,340 on $123,660). Payments are tied to milestones:

Milestone	Share	Amount
M0 kickoff	10%	$17,000
M1 baseline report (week 4)	20%	$33,000
M2 gate (week 8)	20%	$33,000
M3 pilot live (week 16)	30%	$50,000
M4 acceptance (week 20): $16,000 on delivery, $16,000 when the measured result meets the agreed threshold	20%	$32,000

The client can stop at the gate having paid $83,000, and keeps the ledger and benchmark. Optional: fleet utilisation diagnostic (option G), $12,000 fixed, not included above.

8.3 Running cost to the client

Hosting about 600*amonth*(7,200 a year) plus a support and maintenance retainer of 2,000*amonth*(24,000 a year, about 0.2 FTE). Total about $31,200 a year. No per-call AI costs, since the logic is not LLM based.

8.4 Value to the client, with arithmetic

Assumptions: 80,000 t a year purchased (if it is fabricated output, purchases are higher and value rises); overall waste 8.5%, the midpoint of 7 to 10%; net loss per tonne wasted $800 purchase minus $250 scrap = $550.

- Current waste: 80,000 × 8.5% = 6,800 t, costing 6,800 × $550 = $3.74 million a year.
- Improvement is the unknown, so three cases:

Case	Yield gain	Tonnes avoided	Annual value	Less running cost	Net per year	Payback after full rollout
Low	0.4 pts	320 t	$176,000	$31,200	$144,800	165,000 / 144,800 = 1.14 yr (about 14 months)
Base	0.7 pts	560 t	$308,000	$31,200	$276,800	0.60 yr (about 7 months)
High	1.0 pts	800 t	$440,000	$31,200	$408,800	0.40 yr (about 5 months)

Rollout completes around month 5, so payback from kickoff is roughly 19, 12 and 10 months respectively. The 0.7 base case is my estimate from offcut and remnant reuse, not a measurement; the week-8 back-test replaces it with a measured number before the client commits to the second half. Not counted: one-off cash released from surplus stock, avoided engineering-error losses, faster nesting, and any gains in the other areas.

9. Risks

Risk	Likelihood	Impact	Mitigation
Strumis data is incomplete or access is delayed	High	High	Week 4 stop condition; scope manual exports; fixed milestones protect both sides
Nesters resist; they believe manual is better	High	High	Involve the nesting lead from week 1; benchmark on their jobs; suggestions only, they decide
Finance or contracts block cross-project reuse	Medium	High	Ask in week 1; design so in-project reuse works alone; quantify the cost of the rule
Traceability (mill certificates, heat numbers) blocks reuse	Medium	Medium	Filter on traceability class; keep heat number on every suggestion
Drawing revisions invalidate plans mid-project	High	Medium	Revision cause code now; revision tracking as a follow-on
Savings smaller than the base case	Medium	Medium	Price tied to gate; low case still pays back; M4 half-payment tied to results
Projects on hold leave too little new work to pilot	Medium	Medium	Back-test on history; extend pilot rather than slip the milestone
Contact lacks authority; stakeholders not engaged	Medium	High	Steering call every two weeks with head office; written sign-offs
Scope creep into yard and trailers	High	Medium	Written scope; change requests priced separately
Confidential data handled badly	Low	High	Data stays in client tenant; access limited; no public posting

Appendix A. Decisions I made where the notes were unclear

- Currency and region unknown: used USD and assumed the figures in Q8 above.
- The 80,000 tonnes is treated as purchased steel.
- Overall waste of 8.5% is the midpoint of 7 to 10%.
- "Waste" includes unrecovered offcuts; offcuts reused within 90 days count as recovered.
- I did not size options D to F in money because the notes give no data for them.
- I kept the nesting team's claim as a hypothesis, not a finding.

Appendix B. Check on the client's waste figures

Let w be the plate share of tonnes. Overall waste = 22.5% × w + 7.5% × (1 − w) at the midpoints. For overall 7%: w is below 0. For 8.5%: w = 1/15 = 6.7%. For 10%: w = 2.5/15 = 16.7%. So the three numbers fit together only if plate is 0 to about 17% of tonnes, or if they are measured differently. Taking the most favourable ends of the stated ranges (plate 20%, profiles 5%, overall 10%) plate could be at most about one third of tonnes. Either way, plate is a minority of the steel, which is why question 1 comes first.